

Akshata Madavi

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Summary

MS Software Engineering student at SJSU with 6 years of production data engineering and hands-on deep learning shipped in grad school. Trained and deployed a real-time anomaly detection model (XGBoost, 98% accuracy) on time-series sensor data with end-to-end Kafka streaming inference in ~10 seconds on GCP. Fine-tuned LLMs via LoRA (+10 pp accuracy) and built production ML pipelines processing 10M+ records. Brings the rare combination of large-scale data infrastructure depth and modern AI/ML engineering.

Education

San Jose State University, MS in Software Engineering May 2027
Machine Learning, Deep Learning, Recommender Systems, Software Systems Engineering
College of Engineering Pune (COEP), BS in Computer Engineering May 2018

Skills

- **Languages:** Python, SQL
- **Machine Learning:** PyTorch, TensorFlow, scikit-learn, XGBoost, PySpark ML, CNNs; evaluation (accuracy, precision, recall, F1, ROC-AUC)
- **LLMs & GenAI:** Fine-tuning/LoRA (Qwen, LLaMA); RAG (LangChain, LlamaIndex); Vector stores (FAISS, Pinecone); Evaluation (Ragas, Promptfoo); Serving (Vertex AI, HuggingFace)
- **Data & Streaming:** Kafka/KSQL, Spark, BigQuery, Airflow, Pandas, NumPy, MySQL
- **MLOps & Cloud:** MLflow, Docker, GitHub Actions CI/CD, GCP (Vertex AI, BigQuery, GCS), AWS

Projects

VitalWatch – Real-Time Sensor Anomaly Detection (AI-Catalyst Hackathon 2025)

PyTorch, XGBoost, Kafka/KSQL, GCP Vertex AI, BigQuery, Twilio, Python

- Trained XGBoost on multi-channel time-series sensor data achieving **98% anomaly detection accuracy**; deployed on GCP Vertex AI.
- Built Kafka/KSQL streaming pipeline enriching flagged events via LLM with BigQuery data; delivered end-to-end alerts in **~10 seconds**.

Personal Finance Auditor – Agentic Text-to-SQL Assistant

PydanticAI, Gemini, SQLite, sqlglot, Streamlit, Logfire, Python

- Built a multi-agent pipeline: parallel LLM agents categorize bank statement merchants at ingest; a conversational agent answers plain-English spending questions via a Text-to-SQL tool loop against SQLite.
- Enforced two-layer SQL safety guardrail (sqlglot parse-level SELECT validation + read-only SQLite connection); instrumented all agent runs with Logfire traces (LLM calls, tool calls, latency, tokens); deployed via Streamlit UI.

Enterprise AI Governance Platform – LLM Fine-tuning & Monitoring

PyTorch, HuggingFace, LoRA, LangChain, FAISS, MLflow, AWS, Docker

- Fine-tuned Qwen 0.6B via LoRA; tracked experiments with MLflow improving accuracy from **87% to 97% (+10 pp)**.
- Designed LangChain + FAISS retrieval pipelines for failure pattern analysis; containerized with Docker for reproducible training and deployment.

Work Experience

Senior Data Engineer, LTIMindtree – Pune, India

May 2023 – Mar 2025

- Engineered near real-time GCP/BigQuery data pipelines ingesting **~10M records** from enterprise sources, building ML-ready feature tables for downstream AI/ML models at production scale.
- Redesigned a multi-source ETL service (~30 GB) using parallel Spark jobs, cutting ingestion from **1 week to under 10 minutes**; served as the primary feature store for ML workloads.

Data Engineer, AI Adventures – Pune, India

Aug 2021 – Apr 2023

- Led migration of course data from Moodle to MySQL; A/B testing and retention analysis drove a **20% revenue increase**.

Data Engineer, STL (Sterlite Technologies) – Pune, India

Jun 2018 – May 2021

- Built near real-time GCP pipelines migrating Oracle, SAP, and Salesforce to BigQuery (~1 TB); applied DOE-based statistical analysis for manufacturing scrap detection and process optimization.
- Developed automated anomaly detection checks over sensor and telemetry outputs, reducing manual review time and catching production anomalies earlier.